

6	(a) (i) peak voltage = 4.0 V	A1	[1]
	(ii) r.m.s. voltage ($= 4.0/\sqrt{2}$) = 2.8 V	A1	[1]
	(iii) period $T = 20$ ms frequency = $1 / (20 \times 10^{-3})$ frequency = 50 Hz	M1 M1 A0	[2]
	(b) (i) change = $4.0 - 2.4 = 1.6$ V	A1	[1]
	(ii) $\Delta Q = C\Delta V$ or $Q = CV$ $= 5.0 \times 10^{-6} \times 1.6 = 8.0 \times 10^{-6}$ C	C1 A1	[2]
	(iii) discharge time = 7 ms current = $(8.0 \times 10^{-6}) / (7.0 \times 10^{-3})$ $= 1.1(4) \times 10^{-3}$ A	C1 M1 A0	[2]
	(c) average p.d. = 3.2 V resistance = $3.2 / (1.1 \times 10^{-3})$ $= 2900 \Omega$ (allow 2800 Ω)	C1 A1	[2]

7 (a) sketch: concentric circles (minimum of 2 circles)

M1